
LateAct: Commitment-Aware Rollback for Asynchronous Control in Interactive Video World Models

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Abstract

Interactive video world models are usually evaluated with an action fixed before generation, yet deployed controllers can change actions while a few-step diffusion block is already being denoised. We show that this timing matters: action-conditioned generation can cross an *operational denoising-time commitment boundary*, after which direct late binding retains the old action despite using the new action in remaining evaluations. The observed transition is steep at the few-step sampler’s granularity; we do not assume a universal discrete boundary. We introduce LATEACT, a training-free serving policy that checkpoints the latest pre-commitment runtime state and recomputes only the necessary denoising suffix. On Matrix-Game 2.0 mouse-yaw, rollback raises median normalized new-action response from 0.218 to 0.997. In a frozen asynchronous evaluation on 32 fresh contexts and 1,024 paired arrivals, LATEACT matches full restart within 0.10 response on all 676 late arrivals, uses zero rollback on all 348 early arrivals, and reduces mean late-arrival action-to-pixel latency by 11.93% with a compact 339,360-byte exact checkpoint. A frozen confirmatory yaw study on minWM Wan2.1 Action2V passes 16/16 direction-scene gates, independently replicating the commitment mechanism; we do not test rollback or latency on minWM. Keyboard controls share the same boundary and recover directional response, but fail a pre-specified trajectory-fidelity safeguard (future-SSIM $0.6726 < 0.90$). Thus LATEACT validates commitment-aware rollback for Matrix-Game mouse-yaw, while cross-model evidence supports the underlying mechanism rather than universal policy efficacy.

1 Introduction

World models promise inexpensive, inspectable rollouts for planning and control [Ha and Schmidhuber, 2018, Hafner et al., 2023]. Recent generative interactive environments can produce action-conditioned pixels [Bruce et al., 2024, Valevski et al., 2025, Alonso et al., 2024]; few-step causal video diffusion further moves these systems toward real-time streaming [He et al., 2025, Zhao et al., 2026]. This progress exposes a serving question that standard offline evaluations omit: *what happens when a new action arrives while the current video block is still being denoised?*

A server has three unsatisfactory default choices. It can finish the current block under the old action, producing stale control feedback; inject the new action into the remaining denoising evaluations, which is cheap but may be too late; or restart the block, which recovers control but discards all completed work. The middle option implicitly assumes that action conditioning remains equally effective throughout the sampler. Our controlled interventions show that this assumption is false in

35 two open few-step autoregressive video world models. The generated branch can become committed
36 to the old action well before the final denoising evaluation.

37 This is not the claim that early diffusion steps matter in general. Prior editing and sampling work
38 already shows that the effect of conditioning can depend on denoising time. Our question is opera-
39 tional: under identical rollout state and noise, after how many evaluations does substituting a newly
40 arriving action cease to recover the new-action outcome? The contribution is the chain *measurement*
41 $\rightarrow$ *serving boundary* $\rightarrow$ *exact suffix rollback*: we measure action recoverability, turn it into a rollback
42 decision, and audit the dependencies needed to restore the corresponding runtime state.

43 We turn this observation into LATEACT, a commitment-aware rollback policy. LATEACT measures
44 the latest sampler state from which changing the action still recovers the new-action oracle, check-
45 points exactly that state, and, for late arrivals, restores it and recomputes only the committed suffix.
46 Early arrivals bind directly and pay no rollback. The key systems result is that the necessary Matrix-
47 Game state is not the full live KV allocation: because current-block cache slices are overwritten
48 before use, an exact checkpoint contains only the entering-NFE2 latent and defensive cache indices
49 (339,360 bytes).

50 Our contributions are:

- 51 • a controlled, noise-coupled action-switch intervention that identifies an operational recov-
52 erability boundary separately from ordinary video quality;
- 53 • LATEACT, an asynchronous serving policy whose Matrix-Game mouse-yaw response is
54 near full restart with less recomputation and 11.93% lower mean late-arrival latency;
- 55 • a failure-preserving scope analysis: minWM yaw independently replicates the commitment
56 mechanism, whereas minWM lateral control fails its original oracle-identifiability gate and
57 keyboard rollback fails trajectory fidelity.

58 The defensible claim is intentionally asymmetric: LATEACT’s rollback efficacy is validated on
59 Matrix-Game, while denoising-time commitment independently replicates on minWM under a
60 frozen confirmatory yaw protocol.

61 2 Related work

62 **Interactive video world models.** Learned world models range from compact latent dynamics for
63 control [Ha and Schmidhuber, 2018, Hafner et al., 2023] to pixel-generative simulators. Genie
64 learns controllable environments from videos [Bruce et al., 2024]; GameNGen and DIAMOND
65 demonstrate diffusion-based interactive game simulation [Valevski et al., 2025, Alonso et al., 2024].
66 Interactive video generation has also been studied through action conditioning, autoregression, and
67 explicit long-term memory [Chen et al., 2025]. These works primarily ask whether a fixed action
68 sequence controls a rollout. We instead intervene on *when* within a single denoising block the action
69 becomes available.

70 **Time-varying conditioning and editing.** Diffusion models iteratively refine noise [Ho et al.,
71 2020]; distribution matching distillation can compress this process to few steps [Yin et al., 2024].
72 Diffusion editing changes a source at an intermediate noise level [Meng et al., 2022], while Prompt-
73 to-Prompt injects attention maps over selected denoising steps [Hertz et al., 2023]. Conditioning
74 can also be scheduled during sampling, as in condition-annealed sampling [Sadat et al., 2024].
75 These works establish that *when* and *how* conditioning enters diffusion can affect the result; they
76 target editing, fidelity–diversity trade-offs, or semantic control rather than an action that arrives
77 asynchronously during an already executing world-model block.

78 **Few-step generation and inference reuse.** Self Forcing combines few-step generation with
79 causal autoregression and rolling KV state [Huang et al., 2025]. Matrix-Game 2.0 uses a three-
80 evaluation causal generator with mouse and keyboard control [He et al., 2025]. minWM provides a
81 four-step Action2V pipeline over Wan2.1 [Zhao et al., 2026, Wan Team, 2025]. Their speed makes
82 in-flight action arrival practically relevant. Feature-caching methods such as DeepCache reuse tem-
83 porally redundant diffusion computation [Ma et al., 2024]. In contrast, LATEACT does not approx-
84 imate a denoiser or cache features for unconditional speedup: it switches an action under coupled

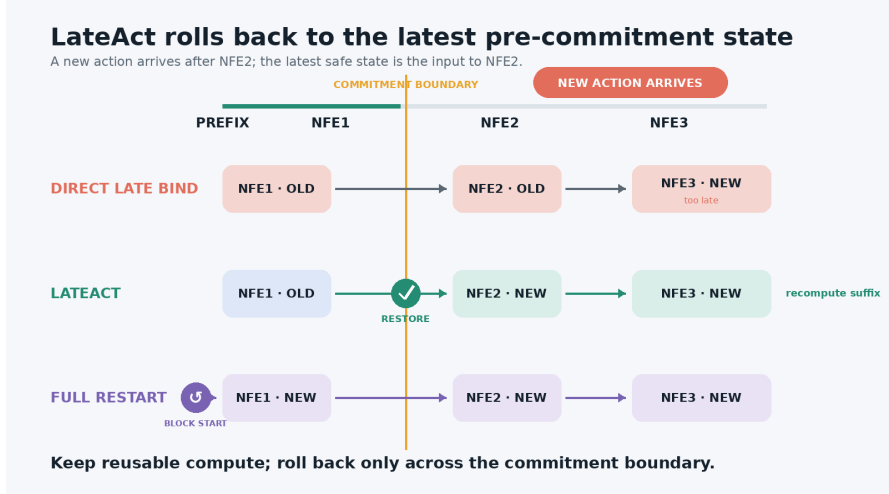


Figure 1: **LATEACT restores the latest pre-commitment state.** With a new action arriving after NFE2, direct binding changes only NFE3 and remains committed to OLD. Full restart recomputes all three evaluations. LATEACT restores the input to NFE2 and recomputes NFE2–NFE3 under NEW, preserving reusable NFE1 work.

85 state/noise, measures when the new branch remains recoverable, and uses that boundary to select
 86 an exact checkpoint and replay suffix. The logical policy resembles checkpoint/replay serving, but
 87 checkpoint contents follow an architecture-specific dependency audit.

88 3 Asynchronous action commitment

89 Consider one autoregressive video block generated by K denoising network evaluations. A prefix
 90 state h , initial noisy latent z_K , materialized future noise ϵ , sampler $\mathcal{S}$, and action a define a deter-
 91 ministic rollout

$$y = F(h, z_K, \epsilon, \mathcal{S}; a). \quad (1)$$

92 Let a_o and a_n be opposite old and new actions. The switched rollout y_s uses a_o for the first s
 93 evaluations and a_n for the remaining $K - s$. Thus $y_0 = y_n$ is the new-action oracle and $y_K = y_o$
 94 the old-action oracle. Given a signed action-response evaluator $g(\cdot)$, we define

$$R(s) = \frac{g(y_s) - g(y_o)}{g(y_n) - g(y_o)}, \quad (2)$$

95 without clipping. $R = 1$ denotes new-oracle behavior and $R = 0$ old-oracle behavior. We evalu-
 96 ate both switch directions and require separated oracles, metric-sign agreement, adequate tracking
 97 support, exact stochastic coupling, and valid videos before interpreting R .

98 An *operational commitment boundary* lies between adjacent evaluations where R undergoes a ma-
 99 terially steep drop at the available sampling granularity. The latest safe state is the last state from
 100 which applying a_n retains near-oracle response. This is a controlled causal intervention on the model
 101 rollout, not a claim of physical causal correctness. The boundary is a property of the model, action
 102 family, sampler, metric, and evaluation granularity; it is not assumed universal or infinitely sharp.

103 4 LATEACT

104 **Policies.** Figure 1 contrasts four serving policies. NEXT-BLOCK finishes the current block under
 105 OLD and applies NEW only to the next block. DIRECT uses NEW at the next available evaluation
 106 without restoring state. RESTART discards current progress and recomputes all K evaluations under
 107 NEW. LATEACT directly binds arrivals that occur before commitment; otherwise it restores the
 108 latest safe state and replays only the necessary suffix.

109 **Compact exact checkpoint.** In Matrix-Game, the latest safe point is the input to NFE2. Static
 110 and runtime tracing show that NFE2 overwrites every current-block visual and action KV posi-
 111 tion before reading it. Historical-prefix and cross-attention KV remain immutable. The imple-
 112 mented checkpoint therefore stores the BF16 entering-NFE2 latent (337,920 bytes) and defensive
 113 visual/mouse/keyboard cache indices (1,440 bytes), totaling 339,360 bytes. Scheduler state is state-
 114 less for the fixed tensors, and random noise is materialized before branching. Independent replay
 115 audits require bit-exact equality between restored and freshly generated OLD-NFE1/NEW-NFE2-
 116 NFE3 trajectories.

117 **Arrival scheduling.** An action becomes usable after the currently executing evaluation completes.
 118 An arrival during NFE1 binds NEW for NFE2–NFE3 without restore. An arrival during NFE2
 119 or NFE3 restores the entering-NFE2 checkpoint and replays NFE2–NFE3. This fixed rule is cali-
 120 brated once on disjoint commitment scenes; the asynchronous evaluation never refits the boundary.

121 **Algorithm 1: calibrate and serve with LATEACT.** **Calibrate:** generate y_s for $s = 0:K$ with coupled
 state/noise; compute $R(s)$ and retain the state at the latest switch s^* passing the frozen gates. **Serve:** bind
 a_n directly if actionable by s^* ; otherwise restore s^* and replay only $s^*+1:K$ under a_n . **Audit:** trace
 mutable dependencies read by the suffix to derive the payload, then verify exact replay.

122 The policy above is architecture-independent; the dependency audit is not. In particular, neither
 123 Matrix-Game’s payload nor its 339,360-byte size is assumed to transfer to another model.

124 5 Experimental design

125 **Substrates and actions.** The primary substrate is the open Matrix-Game 2.0 causal generator [He
 126 et al., 2025], evaluated in BF16 with three denoising evaluations per block. The primary action is
 127 opposite mouse-yaw; keyboard A/D is a pre-specified limitation study. Mechanism replication uses
 128 the pinned minWM Wan2.1 Action2V four-step DMD checkpoint [Zhao et al., 2026].

129 **Splits.** Matrix-Game commitment calibration uses eight official images and both yaw directions
 130 (16 pairs). Minimal rollback uses the same frozen pairs only after the boundary gate passes. The
 131 final serving test uses 32 fresh stochastic contexts: eight disjoint base images, four deterministic
 132 variants per base, two directions per variant, and 16 deterministic stratified arrivals per direction
 133 (64 direction cells and 1,024 paired arrivals). Statistical inference clusters by the eight independent
 134 base images; the 64 direction cells remain descriptive quality units. Keyboard calibration uses 10
 135 fresh stochastic contexts (20 directions), and its serving confirmation uses 24 disjoint contexts (48
 136 directions).

137 The original minWM study fixes official prompts 0–7, seeds 41000–41007, and native lateral a*3
 138 versus d*3. Its formal gate fails and is preserved. After that identifiability audit, exactly one follow-
 139 up was frozen at commit f1e2bbc before its outputs were generated. It uses the same model, evaluator,
 140 thresholds, and gate with stronger native yaw j*3 versus l*3, fresh consecutive official prompts
 141 8–15, seeds 41008–41015, both directions, and all switch positions $s = 0, \dots, 4$. No scene is
 142 screened or replaced, and rollback is prohibited.

143 **Coupling and metrics.** Every branch shares prefix, initial latent, explicit re-noising tensors, sam-
 144 pler, weights, historical state, and non-action conditioning; only the action tensor at designated
 145 evaluations differs. Same-action controls must produce exact latent hashes. Signed horizontal mo-
 146 tion is measured with forward/backward-consistent Lucas–Kanade tracks [Lucas and Kanade, 1981]
 147 and a RANSAC partial-affine fit [Fischler and Bolles, 1981]; the second estimate serves as a com-
 148plementary sign-consistency check. Keyboard uses the same construction but an action-appropriate
 149 lateral scene-translation score rather than the mouse-yaw score. Quality uses future-frame SSIM
 150 [Wang et al., 2004], temporal consistency, boundary jump, and finite/nonconstant decode checks.
 151 Action-to-pixel latency includes checkpoint operations and VAE decoding.

152 **Frozen gates.** Matrix-Game requires all 16 directions to be evaluator-valid and a steep, bounded,
 153 monotone response transition. minWM requires an oracle gap of at least 8 pixels, at least $10\times$ the
 154 same-action floor, at least 20 median inliers, and affine/LK sign agreement. Confirmation requires

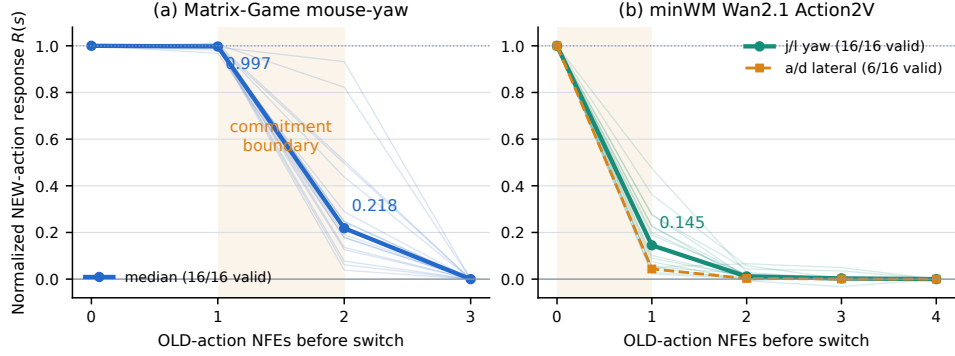


Figure 2: **Action response shows a steep commitment transition at few-step sampling granularity.** Thin lines are direction-scene curves; bold lines are medians. (a) Matrix-Game mouse-yaw retains NEW after one old-action evaluation (0.997) but falls to 0.218 after two (16/16 valid). (b) The frozen minWM yaw follow-up passes 16/16 pairs and drops to 0.145 after one of four old-action evaluations. The original minWM lateral curve is descriptively similar but remains a failed replication because only 6/16 pairs meet the unchanged oracle-separation gate.

at least 12/16 valid monotone bounded pairs, six per direction, and an adjacent median drop of 0.15. The thresholds serve distinct safeguards: the 8-pixel gap enforces absolute action distinguishability, the $10\times$ same-action floor bounds repeatability noise, the 20-inlier requirement supports estimator stability, and the 0.15 adjacent-drop gate requires a materially steep transition. Separately, the 0.10 response tolerance defines operational near-restart equivalence, while median SSIM ≥ 0.90 serves as the frozen trajectory-fidelity safeguard. Appendix A.4 gives the full rationale. All precede broad inference. Experiments run on two RTX 3090 GPUs for Matrix-Game and one RTX 3090 for minWM; no training or weight modification is performed.

6 Results

6.1 Denoising-time commitment

Matrix-Game. All 16 mouse-yaw pairs are valid, monotone, and bounded. Median $R(s)$ for $s = 0, 1, 2, 3$ is $[1.000, 0.997, 0.218, 0.000]$, with a paired median drop of 0.774 from one to two old-action evaluations. This descriptive 16/16 result places the latest safe checkpoint at the input to NFE2.

Independent minWM mechanism chronology. The original eight-scene lateral a/d study passes oracle separation on only 6/16 pairs (three per direction), below the frozen 12/16 requirement. Its commitment-like curve is descriptive only; action separation, not the implementation audits, causes the identifiability failure.

Following this failed identifiability audit, the single pre-specified frozen j/l yaw follow-up uses fresh prompts and unchanged evaluation, thresholds, and success gate. It passes 16/16 pairs (8/8 per direction); all validity and coupling audits pass, and median $R(s)$ is $[1.000, 0.145, 0.011, 0.003, 0.000]$. This replicates the *commitment mechanism* on a second model, not LATEACT rollback or latency; full protocol and failure chronology appear in Appendix A.6.

6.2 Rollback recovers the new action

On the 16 Matrix-Game calibration directions, direct binding from the post-NFE2 state has median response 0.218, while restoring the input to NFE2 yields 0.997; full restart is 1.000. Minimal rollback improves all 16/16 pairs and is within 0.10 of restart on all 16. Its median future-SSIM to the new oracle is 0.9026, passing the frozen median-level 0.90 safeguard. A representative coupled branch is retained in Appendix A.4 rather than used as primary evidence.

Table 1: **Matrix-Game mouse-yaw serving results over 1,024 paired arrivals.** Response and latency are descriptive medians. For NEXT-BLOCK, response is for the current block, while latency ends when NEW first affects pixels in the following block. LATEACT reaches near-restart normalized response with two post-arrival evaluations rather than three.

Policy	Response	Latency (s)	Post-arrival NFEs	Redone NFEs
NEXT-BLOCK	0.0000	5.1554	4	0
DIRECT	0.1605	2.2925	2	0
RESTART	1.0000	2.5900	3	2
LATEACT	0.9999	2.2843	2	1

184 6.3 Frozen asynchronous serving evaluation

185 Table 1 reports the disjoint serving test. For all 676 arrivals during NFE2 or NFE3, LATEACT is
 186 within 0.10 of full restart, redoes one fewer evaluation, and saves 11.93% mean late-arrival latency.
 187 For all 348 NFE1 arrivals, direct binding and LATEACT are the same trajectory, with zero restore,
 188 discard, or redone evaluations.

189 Inference uses the eight independent base images, not the 64 derived variant/direction cells. Aver-
 190 aging the actual 676 late-arrival rows within this hierarchy, LATEACT minus DIRECT response is
 191 0.9091 mean and 0.9304 median across bases (cluster-bootstrap 95% CI [0.8598, 0.9541]); all 8/8
 192 base-image effects are positive. Full-restart minus LATEACT latency saving is 0.3118 s mean and
 193 0.3137 s median (95% CI [0.3055, 0.3175]), again positive for 8/8 bases. The corresponding exact
 194 one-sided Wilcoxon value is $p = 0.0039$ for each effect.

195 Mouse-yaw trajectory quality is broadly stable but not uniform. Over the 64 descriptive scene-
 196 variant/direction comparisons, future-SSIM has median 0.9223 and p05 0.8397; 60/64 are at least
 197 0.85. The frozen ≥ 0.90 requirement is a median-level safeguard, not a per-direction threshold.
 198 Temporal and finite/nonconstant decode safeguards pass 64/64; Appendix A.4 discloses the full tail,
 199 including the unfavorable 0.7184 minimum.

200 **State and runtime.** The compact exact checkpoint is 339,360 bytes ($1,913\times$ smaller; 1,432
 201 bytes above the logical minimum), with 1.456/0.878 ms save/restore. All 80 replays pass exact
 202 state/scheduler/RNG audits (peak 8.908 GiB).

203 6.4 Keyboard: shared boundary, failed fidelity

204 Frozen action-appropriate calibration corrects the initial mouse-yaw score and finds the *same* bound-
 205 ary: $R(1) = 0.9957$, $R(2) = 0.1240$. Across 24 contexts, rollback reaches median response 1.0188
 206 and is near restart for all 512 late arrivals, but future-SSIM is 0.6726, below the frozen 0.90 median
 207 gate (temporal 48/48). Shared commitment and direction do not imply action-general fidelity.

208 7 Discussion and limitations

209 Matrix-Game mouse-yaw supports compact exact rollback with near-restart response and lower la-
 210 tency. minWM supports mechanism replication only; a/d remains failed and keyboard precludes
 211 action-general fidelity.

212 **Deployment scope.** Full efficacy is limited to Matrix-Game mouse-yaw; minWM tests commit-
 213 ment, not rollback/latency. Arrivals are stratified, not production-sampled; latency is about 2.28 s;
 214 and the boundary is calibrated, not predicted online. No task-level physical-control objective or real
 215 robot is evaluated.

216 8 Conclusion

217 LATEACT turns a measured action-recoverability boundary into exact suffix replay, yielding near-
 218 restart mouse-yaw response with safeguards and lower Matrix-Game latency. minWM mechanism
 219 replication and the negative results delimit this claim.

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275 A Frozen protocols and additional results

276 A.1 Matrix-Game commitment and rollback gates

277 **Commitment calibration.** The calibration set contains official universal images 0000–0007 and
 278 both mouse-left↔mouse-right directions. For each of the 16 direction-scene pairs, the runner gen-
 279 erates the NEW and OLD oracles and the two interior switch points. Prefix, prompt-free image
 280 conditioning, initial current-block latent, future noise tensors, weights, sampler, precision, cache
 281 history, and non-action conditioning are exact across branches. The intended action tensor is the
 282 only difference. All 16 pairs pass oracle distinction, tracking support, monotonicity, boundedness,
 283 temporal quality, and latent coupling.

Table 2: Frozen Matrix-Game commitment curve. Values are medians over 16 valid direction-scene pairs.

OLD-action NFEs s	0	1	2	3
Normalized response $R(s)$	1.0000	0.9971	0.2182	0.0000

284 **Gate 1 rollback.** The 16 calibration pairs compare DIRECT, exact rollback of the minimal suffix,
 285 and full restart with identical stochastic inputs. Minimal rollback is within 0.10 of restart on 16/16
 286 pairs and improves over direct binding on 16/16. Median responses are 0.2182/0.9971/1.0000 for
 287 direct/minimal/restart, and minimal rollback has median future-SSIM 0.9026.

288 A.2 Exact Matrix-Game runtime state

289 Matrix-Game routes causal self-attention KV insertion through a single cache operation immediately
 290 before attention reads. Runtime tracing shows that every current-block visual and active action KV
 291 position is overwritten before use at NFE2. Historical prefix KV and cross-attention KV therefore
 292 remain live by reference rather than being copied.

Table 3: Exact checkpoint storage. The final implementation removes overwritten state, not numerical information required by replay.

Payload	Bytes
Entering-NFE2 BF16 latent [1, 16, 3, 44, 80]	337,920
Defensive visual/mouse/keyboard cache indices	1,440
Implemented exact checkpoint	339,360
Mathematical minimum (latent plus step)	337,928
Initial exact implementation	649,330,080

293 The final checkpoint is $1,913\times$ smaller than the initial implementation and 1,432 bytes (0.424%)
 294 above the measured logical minimum. Median save/restore times are 1.456/0.878 ms. The full live
 295 per-fork cache allocation is 1,670,124,960 bytes but is not duplicated by LATEACT.

296 A.3 Asynchronous serving details

297 The primary serving set uses official images 0008–0015 with four deterministic prefix-noise variants
 298 each. All 32 contexts and both yaw directions are retained. Each direction receives 16 deterministic
 299 stratified-uniform arrival times over the wall-clock support of the three old-action evaluations, using
 300 one frozen seed family. The resulting classes contain 348 NFE1 arrivals, 338 NFE2 arrivals, and
 301 338 NFE3 arrivals.

302 The serving hierarchy has eight independent base images, four deterministic variants per base, two
 303 directions per variant, and 16 arrivals per direction. Late arrivals have `inflight_nfe` ≥ 2 , re-
 304 taining 676 of 1,024 rows. Inference averages late rows within direction and the eight derived
 305 variant/direction cells within each base, then treats the eight base images as clusters. The late-
 306 arrival LATEACT-minus-DIRECT response effect has mean 0.909136, median 0.930385, and base-
 307 cluster bootstrap 95% CI [0.859769, 0.954093]; all 8/8 base effects are positive (exact one-/two-
 308 sided Wilcoxon $p = 0.00390625/0.0078125$). The fixed per-direction `lateact-direct_late`
 309 branch contrast is a distinct descriptive $n = 64$ estimand (mean 0.815919), not the mean over the
 310 676 asynchronous late rows.

311 Full-restart-minus-LATEACT latency saving across the eight base clusters has mean 0.311789 s, me-
 312 dian 0.313654 s, and 95% CI [0.305519, 0.317523] s; 8/8 are positive with the same exact Wilcoxon
 313 values. The frozen mean saving fraction is 11.93%. Every late case is within 0.10 normalized
 314 response of restart and uses fewer redone evaluations. All 348 early cases use exactly zero rollback.

315 A.4 Mouse-yaw trajectory-quality tail

316 The 64 scene-variant/direction comparisons are descriptive quality units. The future-SSIM distri-
 317 bution has mean 0.915660, median 0.922305, p05 0.839679, and minimum 0.718375. Counts at
 318 least 0.90/0.85/0.80 are 49/64, 60/64, and 63/64; equivalently 15/64, 4/64, and 1/64 fall below those
 319 values. Sub-0.90 values occur in four base images; base 0013 contains 6/8 and the sole value be-
 320 low 0.80. Thus mouse-yaw trajectory quality is broadly stable but not uniform. The frozen SSIM
 321 ≥ 0.90 gate is a median-level safeguard, not a per-example threshold, and no tail cutoff is introduced
 322 retrospectively.

323 Temporal difference (LATEACT minus NEW) has median +0.001691 and range
 324 $[-0.007878, 0.018057]$; temporal and finite/nonconstant decode safeguards pass 64/64.

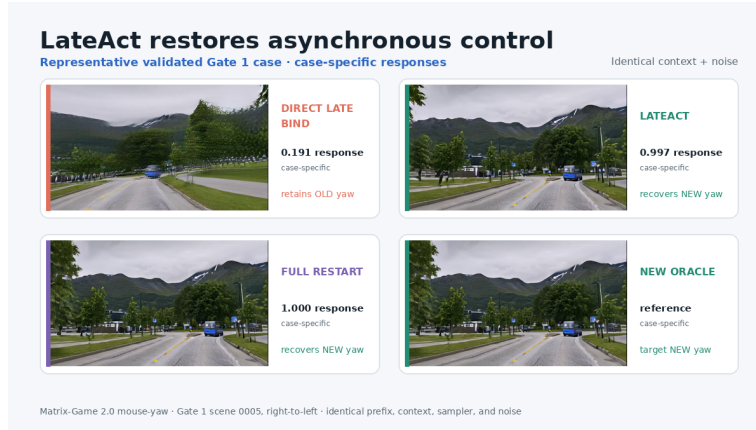


Figure 3: Representative coupled Matrix-Game mouse-yaw branch. Direct late binding retains the old yaw (response 0.191), while LATEACT (0.997) tracks full restart (1.000) in normalized action response. This favorable example is not the quality-tail evidence and was not used to set thresholds.

325 A.5 Gate rationale

326 The frozen numerical cutoffs are safeguards against specific invalidating conditions, not uniquely
 327 privileged constants. The 8-pixel oracle gap demands minimum absolute action separation; the
 328 $10\times$ same-action floor demands separation well above repeatability noise; and at least 20 motion

inliers protects the affine estimate from sparse tracking. A 0.10 normalized-response gap is the operational near-restart tolerance. Median future-SSIM ≥ 0.90 protects trajectory fidelity without requiring every example to exceed 0.90. Finally, a median adjacent response drop of at least 0.15 distinguishes a materially steep transition at the available NFE granularity from a flat or noisy curve.

A.6 minWM frozen chronology

Original lateral control. The original minWM protocol fixes prompts 0–7, seeds 41000–41007, actions $a \times 3/d \times 3$, both directions, and $s = 0, \dots, 4$. The same-action decoded floor is 0.0776 pixels. Oracle affine gaps have median 6.851 pixels and range 1.169–12.767; only three scenes in both directions exceed the fixed 8-pixel threshold, yielding 6/16 valid pairs. The formal replication verdict is failure even though all 16 raw curves are monotone and bounded and show a median first drop of 0.9566.

Frozen yaw confirmation. After the lateral identifiability failure was established, exactly one bounded follow-up protocol was frozen at commit f1e2bbc before confirmatory outputs were generated. It fixes fresh consecutive official prompts 8–15, seeds 41008–41015, native $j \times 3/l \times 3$ yaw, both switch directions, the unchanged evaluator, thresholds, and success gate. Prompt screening/replacement, alternative evaluators or actions, and rollback are prohibited. This is a pre-specified frozen follow-up after a failed identifiability audit; the adaptive chronology is explicit, and the protocol was fixed before confirmatory execution.

Table 4: Per-scene minWM yaw oracle separation. The signed value reverses between directions; absolute gap and inlier support are shared.

Scene	Absolute gap (px)	Median oracle inliers	Valid directions
08	35.359	177.0	2/2
09	14.002	833.5	2/2
10	13.982	690.5	2/2
11	35.345	469.5	2/2
12	10.238	298.5	2/2
13	19.007	246.0	2/2
14	19.035	631.5	2/2
15	45.441	399.5	2/2

The same-action control ranges are 0.3892 and 0.6580 pixels; the latter is the frozen repeatability floor, giving a $10\times$ requirement of 6.5801 pixels. All 16 pairs exceed both this value and 8 pixels. Affine/LK signs agree 16/16, all median inlier counts exceed 20, all videos are valid, and all 16 curves are monotone and bounded. Exact branch-noise hashes, historical/cross-sampled state, and four-evaluation counts pass on all eight scenes. Same-action latent hashes are exact on both control prompts.

A.7 Keyboard chronology

The initial eight-context keyboard transfer used the mouse-yaw score and reported asymmetric response. The subsequent diagnostic froze an action-appropriate signed lateral translation evaluator before generating broad results. On 10 calibration contexts and 20 directions, all directions are valid; median response is 0.9957 at $s = 1$ and 0.1240 at $s = 2$, placing the latest safe state after NFE1, exactly as for mouse-yaw.

The confirmation uses 24 disjoint stochastic contexts, 48 directions, and 16 arrivals per direction. Mouse-boundary and action-specific policies are the same algorithm and are bit-exact on 768/768 records. Median response is 1.0188; all 512 late arrivals are within 0.10 of restart and save 0.3105 s median. Nevertheless, median future-SSIM to NEW is 0.6726 (range 0.5989–0.9139), failing the frozen median-level 0.90 threshold. Temporal consistency passes 48/48. The videos are coherent but follow a different exact trajectory.

B Reproducibility and audit details

No model is trained or fine-tuned. The experiments use released Matrix-Game 2.0 and minWM Wan2.1 Action2V checkpoints, BF16 inference, deterministic materialized future noise, and frozen YAML protocols. Runners, analyzers, protocol validators, configs, and lightweight reports are retained for an anonymous release; large checkpoints and generated videos are excluded.

For each scientific branch, the runner records hashes of action tensors, initial and re-noising tensors, prefix latents, cache indices, sampled historical cache bytes, cross-attention state, and output latents. Analyzer guards reject prompt substitution, action mismatch, threshold drift, output overwrite, insufficient oracle separation, evaluator-sign disagreement, non-finite videos, and failed same-action equality.

Matrix-Game generation uses two NVIDIA RTX 3090 GPUs. The primary serving runner measures 955.60 synchronized GPU-seconds across 64 scene-directions (6.21 s generation and 8.72 s decode per bundle). minWM uses one RTX 3090; 74 confirmatory branches average 6.156 s generation and 7.417 s decode (1,004.42 timed GPU-seconds total), with 19,340,754,432 peak allocated bytes.

C Existing assets and broader impact

The work evaluates released Matrix-Game 2.0 and minWM implementations and checkpoints and cites their corresponding papers. The Matrix-Game 2.0 source release identifies an MIT license; minWM is released under Apache-2.0. Model checkpoints and official evaluation assets are used under their respective upstream terms and are not redistributed in the submission.

Commitment-aware rollback can reduce wasted inference and make interactive world-model feedback more responsive, which may benefit embodied-agent simulation and human-in-the-loop control. The same capability can also make synthetic environments more persuasive despite incorrect dynamics. A particularly relevant failure mode is false confidence: recovering an action-direction metric does not guarantee recovery of the exact intended trajectory, as demonstrated by the keyboard SSIM failure. These systems should therefore not control safety-critical physical hardware solely from the evaluated pixel or motion metrics. Deployment would require task-level state validation, uncertainty monitoring, safe fallback behavior, and evaluation on the target embodiment.

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